

OPTIMIZING SCHOOL ENTRANCE THROUGHPUT USING AUTOMATED MULTI- SENSOR CLASSIFICATION TO ELIMINATE MANUAL BOTTLENECKS.

Iteration 1

Problem Identification:

How can we automate the detection of medium-sized and large-sized metal objects at the school entrance, ignoring small and non-threatening metal objects?

Decomposition:

- **Amplitude (Strength):** Measures signal peak to ignore "noise" (zippers) and categorize size (Small, Medium, or Large).
- **Duration (Time):** Measures how long the signal lasts to distinguish a solid mass from a quick interference spike.
- **Validation:** Cross-checks both metrics; high strength + sustained time = confirmed large object.

Pattern Recognition:

- **Clutter (Ignore):** Low strength + short time (e.g., keys, coins).
- **Mass Signature:** Consistent, high-density peaks used for object identification.
- **Threat (Alert):** High strength + long duration (e.g., large metallic masses).

Abstraction:

- **Focus (Amplitude):** Tracks only the peak signal strength ("Metal Magnitude") against the baseline.
- **Ignore (Complexity):** Filters out shape, metal type, and time spent in the gate to simplify the decision.

Iteration 2

Problem Identification:

How can we automate the detection of inductive metal items to identify suspicious objects like vapes/e-cigarettes, considering that some objects have a plastic or non-metal casing?

Decomposition:

- **Signature Extraction:** Using "Phase Angles" to identify the specific electronic signature of batteries and coils.
- **Material Classification:** Using "Decay Rates" to tell the difference between high-conductivity metals (aluminum) and low-conductivity metals (steel).
- **Casing Penetration:** Using "Magnetic Transparency" to ignore non-conductive materials (plastic) and see only the internal components.

Pattern Recognition:

- **Electronic Fingerprint:** Uses "Phase Shift" to distinguish the unique electrical signature of a vape from common items like foil or cans.
- **Conductivity vs. Amplitude:** Identifies vapes by their high conductivity (Phase Angle) even when their physical size (Signal Amplitude) is small.

Abstraction:

- **Relevant Information:** Phase Angle (Conductivity) and Inductive Response.
- **Irrelevant Information:** The shape, color, or material of the outer casing (plastic, rubber, or leather).

Iteration 3

Problem Identification:

How can we differentiate between a single large, high-threat object and multiple small, harmless items that create a large total signal? Multiple small items often appear as one large object to standard sensors, leading to unnecessary manual bag checks and bottlenecks.

Decomposition:

- **Spatial Mapping:** Uses the sensor array to locate exactly where metallic mass is concentrated on the person.
- **Peak Isolation:** Analyzes the signal shape to tell the difference between one solid mass (a threat) and multiple scattered points (keys or coins).
- **Threshold Balancing:** Sets a specific density limit that allows small "clusters" to pass while flagging high-density, solid objects.

Pattern Recognition:

- **Solid Mass:** High density (Levels 7-10) concentrated in one spot, signaling a single large object.
- **Bag Clutter:** Low-to-medium density (Levels 1-6) spread out, signaling many small, separate items.
- **Density vs. Magnitude:** Differentiates a solid threat from a bag of change by analyzing the "shape" (scattered vs. concentrated) rather than just the total amount of metal.

Abstraction:

- **Focus (Spatial):** Analyzes metal concentration and the location of signal peaks to find solid objects.
- **Ignore (Physical):** Filters out the total bag weight, non-metallic contents, and branding, as they don't affect the magnetic sensor array.

TECHNIQUES USED IN SOLVING THE PROBLEM

Encapsulation	Created Bag and MetalDetector classes to manage data.
Solution Modeling	Developed a scoring formula based on metal, duration, angle, and density.
Optimization	Used Pandas to handle large datasets efficiently.
Dynamic Thresholding	Categorized threat levels (LOW, MEDIUM, HIGH) based on score averages.